| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 15 | 29.5 or 29.45 to 29.46 | **2** | **M1** for $\frac{60}{360}\times\pi\times7.5^2$ oe |
| 16 | 25 | **2** | **M1** for $x\times\left(1+\frac{6}{100}\right)=26.50$ oe or better |
| 17(a) | 0.1 or $\frac{1}{10}$ | **1** | |
| 17(b) | 90 | **3** | **M2** for $\frac{1}{2}\times10\times2+10\times2+\frac{1}{2}(2+4)\times20$ oe<br>or **M1** for one area calculation or indicated on diagram |
| 18 | 27.15 cao | **3** | **M2** for (9.4 + 0.05) × 2 + 8.2 + 0.05 or better<br>or **M1** for 8.2 + 0.05 or 9.4 + 0.05 or better seen<br>OR<br>**SC2** for answer 25.95<br>or **SC1** for answer 26.85 |
| 19(a) | 61.1 or 61.08 to 61.09... | **3** | **M2** for [$\sin x$ =] $\frac{8\sin100}{9}$ oe or better<br>or **M1** for $\frac{9}{\sin100}=\frac{8}{\sin x}$ oe |
| 19(b) | 11.7 or 11.66 to 11.67 | **3** | **M2** for $\frac{1}{2}\times9\times8\times\sin(180-100-$ *their* **(a)**$)$ oe<br>or **M1** for 180 – 100 – *their* **(a)** |
| 20 | 60 | **3** | **M2** for $4\times\sqrt[3]{\frac{40500}{12}}$ oe<br>or **M1** for $\left(\frac{4}{l}\right)^3=\frac{12}{40500}$ oe<br>or $\sqrt[3]{\frac{40500}{12}}$ oe or $\sqrt[3]{\frac{12}{40500}}$ oe |
| 21(a) | 4 – 2$x$ | **2** | **B1** for 4 or – 2$x$ |
| 21(b) | (2, 10) | **2** | **B1** for $x$-coordinate of 2<br>or **M1** for *their* 4 – 2$x$ = 0 |
| 22(a) | –**a** + **b** | **1** | |